Computational Mental Health Monitoring

CS 8803

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Check-in #1

Feb 4

Who we are!

- Grad students in Analytics Program
- Interests
  - Amit: Technology for social impact
  - Saehee: Applications of SOTA models
- Shepherds: Dr. Nina Dalkner (Clinical Psychologist)
- Industry Partner: Primaface (Mental health tech company)

Source: Team contract

Our topic!

An app that therapists can use to monitor mental health of patients granularly & quantitatively

- Daily check-ins using a mobile device can be a great source of information providing richer insights
- Acoustic analysis (audio) and Linguistic Analysis (text)
- Time series analysis (personalized baseline)

Rough idea of what we want to build with Primaface

Daily Voice Check-In by Patient → **Feature Extraction** → **Mental State Estimation** → **Dashboard Visualization for Therapists**

Patient records spoken responses to guided prompts and/or to an interactional AI (e.g., “How are you feeling today?”).

Voice captures not just words but *acoustic and linguistic cues* related to emotion and psychological state.

Potential LLM: EVI 3 BY Hume AI

Audio is preprocessed (noise removal, segmentation).

Acoustic features (e.g., pitch) are extracted which have been shown to correlate with mental states such as depression and anxiety.

Linguistic features (e.g., negativity) could also be extracted

Machine learning models (from classic classifiers to state-of-the-art transformers) assess risk indicators for depression/anxiety.

Potential Model: Transformer-based NLP models analyze the *content and context*

Results include *confidence scores*, *trend patterns*, and *severity estimations*.

Time-series trends of mood indicators.

Alerts for significant changes or “red-flag” patterns.

Acoustic-based scores (e.g., variability in tone, pauses, energy).

Word cloud or semantic metric summaries derived from text.

Open Design & Research Questions (1/3)

1. Patient Check-In & Data Collection

Pipeline Stage	What is Missing / Unclear
Check-in Frequency	Optimal frequency of check-ins (daily vs multiple times/week vs adaptive)
Prompt Design	What kind of prompts elicit clinically useful signals (open-ended vs structured)
Duration of Speech	Minimum effective recording length for reliable inference
Compliance & Fatigue	How to reduce patient fatigue while maintaining signal quality

2. Feature Extraction & Signal Representation

Pipeline Stage	What is Missing / Unclear
Acoustic Features	Sorting out significant features
Linguistic Features	How to capture semantic and psychological cues
Noise & Variability	Impact of environment, device, accent, and speaking style

Open Design & Research Questions (2/3)

3. Modeling & Inference

Pipeline Stage	What is Missing / Unclear
Model Choice	Which models work best for voice-based mental health signals
Multimodal Fusion	How to combine acoustic + linguistic features effectively
Personalization	General models vs patient-specific baselines
Interpretability	How to explain model outputs to clinicians

4. Dashboard & Clinical Usability

Pipeline Stage	What is Missing / Unclear
Signal Presentation	What metrics are clinically interpretable (raw scores vs trends)
Cognitive Load	Avoid overwhelming therapists with raw data
Trust & Adoption	How to ensure clinicians trust the system

Open Design & Research Questions (3/3)

5. Ethics, Privacy & Deployment

Pipeline Stage	What is Missing / Unclear
Privacy	Handling sensitive voice and mental health data
Bias & Fairness	Model bias across gender, accent, language
Clinical Role	Avoiding over-reliance on AI outputs
Regulatory Readiness	Path toward clinical validation

Roadmap

Phase	Focus Area	What We Will Do
Weeks 1–2	Literature Review	Review speech-based mental health and transformer models; define problem scope, narrow down significant audio/text features from prior work, and build evaluation strategy
Week 3	Data & Check-In Design	Exploratory Data Analysis Decide check-in frequency, prompt design, and assumptions
Week 4	Exploring Models	Checking available models for fine tuning or building from scratch (Audio, Text, Time Series)
Weeks 5	Feature Extraction	Extract acoustic, linguistic, and temporal features from available dataset
Weeks 6–7	Modeling	Train and compare baseline ML, deep learning, and transformer-based models
Week 8	Interpretability	Analyze feature importance and create clinician-interpretable outputs
Weeks 9–10	Dashboard	Design and implement therapist-facing dashboard with trends and alerts
Week 11	Evaluation	Test robustness, analyze failure cases, and assess limitations
Week 12	Finalization	Finalise results and slides, and prepare demo/report

Check-in #2

Feb 19

Recap

About us

Amit & Saehee

MS in Analytics students

Our project

Objective: Building a mechanism to track an individual's mental health through daily/weekly voice check-ins

Technologies: Voice AI, Language AI, Audio processing softwares, Dashboarding

Challenges:

- Baseline highly subjective on individuals
- Time variant results based on check-in time
- Highly experimental and scarce data

Scoping down the problem

Target individuals: GT Students

Target health parameter: Stress

So the questions become** (Illustrative)**:

- What voice parameters can be attributed to stress? Does low arousal means high stress?
- What and how many questions can be asked to students during check-in to perceive their stress levels? Should it be a way one-way conversation or a two-way conversation? What should be the duration of the conversation?
- How do we incentivize students to have these daily check-ins?

Approach

1. Narrow down the features attributing to stress

2. Build interpretable ML models (e.g. regression, decision tree) to predict stress levels

3. Design check-ins based on the selected features and start collecting real data

Some of the identified features after discussions with our shepherds

Acoustic

Prosodic

Pitch instability,
Speech rate, Intensity,
Energy variability,
Hammarberg index

Voice Quality

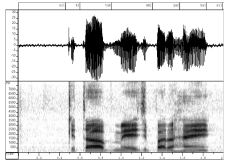
Jitter
Shimmer
Harmonic to noise ratio
Breathiness

Temporal

Utterance duration
Pause count
Pause duration

Spectral

Centroid
Slope
Flux



Semantic

Sentiment

Negative Bias
Positive Affect

Cognitive distortion

Hopelessness language
Overgeneralization
Self-blame
Catastrophizing

Social

Social withdrawal
Connectedness

WHO-5 Proxy

Well being polarity
Energy language

Tools identified to extract the voice features

Feature Categor	Sub-Category	Feature	Tool (Python / Open Source)
Acoustic	Prosodic	Fundamental Frequency (F0) mean	librosa, parselmouth (Praat), openSMILE
		F0 variance / range	parselmouth, openSMILE
		Pitch instability	parselmouth
		Speech rate (syllables/sec proxy via voice	praat-parselmouth, pyannote.audio
		Intensity (RMS energy)	librosa, openSMILE
		Energy variability (std of energy contour)	librosa
		Hammarberg Index	openSMILE
		Alpha Ratio	openSMILE
	Voice Quality	Jitter	parselmouth, openSMILE
		Shimmer	parselmouth, openSMILE
		Harmonics-to-Noise Ratio (HNR)	parselmouth
		Breathiness	openSMILE
		Vocal Tension	openSMILE
	Spectral	MFCC (1–13)	librosa, torchaudio, openSMILE
		Spectral centroid	librosa
		Spectral slope	librosa
		Spectral flatness	librosa
		Spectral flux	librosa
		Formants (F1, F2, F3)	parselmouth (Praat)
		Specific band energy (600–700Hz etc.)	librosa
	Temporal	Utterance duration	librosa, pyannote.audio
		Pause count	pyannote.audio, webrtcvad
		Pause duration (mean/max)	pyannote.audio
		Filled Pauses ("um", "uh")	Whisper + text parsing
		Silent Pauses	webrtcvad
		Proportion of silence	webrtcvad, pyannote.audio
		Fragmented speech (short utterance burs	pyannote.audio

Feature Categor	Sub-Category	Feature	Tool (Python / Open Source)
Semantic	Sentiment	Negativity Bias	transformers (RoBERTa sentiment)
		Positive Affect (resilience answer)	transformers
	Cognitive Distortic	Hopelessness Language	LIWC-style lexicon / custom dictionary
		Overgeneralization ("always", "never")	spaCy
		Catastrophizing	custom lexicon
		Self-blame Language	spaCy / pattern matching
	Cognitive	Rumination (repetition detection)	sentence-transformers + cosine similarity
		"I can't focus" Language	pattern matching
	Social	Social Withdrawal Language	custom lexicon
		Connectedness Score (weekly 0–10)	direct numeric input
	Uncertainty	Hedge Words ("maybe", "I guess")	spaCy
	Functioning	Function Impairment Language	classifier (fine-tuned BERT)
	Sleep	Sleep-related Mentions	keyword extraction
	Stress	Stress Trigger Extraction	zero-shot classification
	Metacognition	"What is on your mind" semantic embedd	sentence-transformers
	WHO-5 Proxy	Well-being Polarity	sentiment model
		Vitality / Energy Language	custom lexicon

Data snapshot

Questionnaire-based scores

1. PHQ:degree of depression severity
2. PCL-C: PTSD symptom severity

DAIC Dataset

~70 hours of audio conversations with clinical patients with PHQ and PCL-C scores

Participant	PHQ8_1_NoInterest	PHQ8_2_Depressed	PHQ8_8_Ps	Depression	PCL-C_1_M	PCL-C_2_D	PCL-C_17_.	PTSD_sever	gender	age	Depression	PTSD_label	split
300	0	0	0	2	2	2	1	25	male	33	0	0	dev
301	0	0	0	3	1	1	1	17	male	39	0	0	dev
302	1	1	0	4	2	2	1	28	male	25	0	0	train
303	0	0	0	0	1	1	1	17	female	41	0	0	train
304	0	1	0	6	1	1	1	20	female	22	0	0	train
305	0	1	0	7	2	1	1	28	male	55	0	0	train
306	0	0	0	0	2	1	1	21	female	62	0	0	dev
307	0	1	0	4	1	1	1	23	female	23	0	0	train
308	2	3	2	22	4	3	3	67	female	40	1	1	train
309	2	1	3	15	5	5	1	74	male	19	1	1	train
310	1	1	0	4	2	1	1	35	male	42	0	0	train
311	3	3	1	21	5	2	4	70	female	62	1	1	train
312	0	0	0	2	1	1	1	18	male	35	0	0	train

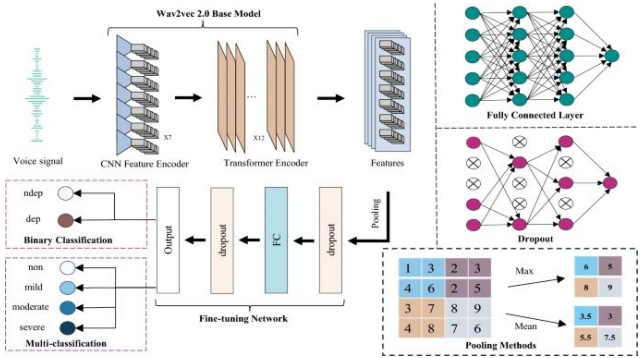
Source: https://dcapswoz.ict.usc.edu/daic-woz-database-download/

Next steps for data preprocessing

1. Convert the PHQ and PCL data into stress labels
2. Extracting the listed features from the audio files
3. Normalizing the features for a population baseline
4. Prepare the data for model ready ingestion

→ Verify which features with significant effect on the y (stress/depression)

Illustration of previously applied model on this dataset



Source: https://www.nature.com/articles/s41598-024-63556-0

Note: Our preprocessing steps might be similar, but for prediction we will use **simpler models (e.g., regression, decision trees) for interpretability**

Thank You